ADDITIVE PUBLIC OPINION SHAPING

Exact Quantum Encoding and Observable Estimation

This deck isolates the additive mainline, the exact strategic encoding result, and the observable-estimation benchmark.

Actors → Mixed Narratives → Observable Payoff

Core message

In the additive setting, the quantum contribution is not a changed game model. It is an exact strategic encoding of the classical ground truth, plus a cleaner route to observable-payoff estimation.

Technical scope: additive GT, exact quantum encoding, residual MLP surrogate, and AE benchmark.

PROBLEM BACKGROUND

Why additive public-opinion shaping is the right reduction.

- Multiple actors intervene in one shared opinion space rather than in isolated channels.
- Each coordinate represents a narrative axis such as economic framing, identity, or social value.
- Actors allocate influence across those coordinates, so strategy is a distributional decision, not a single move.
- Platform redistribution and audience cognition mix narratives across coordinates, which creates strategic coupling.

Additive modeling makes this problem explicit: it turns shared narrative shaping into a state update plus a measurable strategic payoff.

SCENARIO: PUBLIC OPINION SHAPING

- Multiple actors attempt to influence a shared public opinion space
- The state represents a fixed set of narrative dimensions. Each coordinate corresponds to a specific framing axis, such as economic impact, social values.
  All actors operate on the same set of dimensions.
- Each actor allocates influence across these dimensions. This determines how strongly different aspects of an issue are emphasized in public perception.
- these dimensions are mixed by: platforms redistribute attention, narratives interact across topics, and audiences perception mixes different aspects of an issue.
  Where aligned narratives reinforce each other, conflicting narratives weaken or distort each other.

ADDITIVE MAINLINE FORMULATION

Three models share one distribution-based scoring interface.

ClassicalGroundTruthGame

Reference additive dynamics over influence distributions with explicit mixing matrix M and agent-specific maps B_i.

QuantumEncodedGame

Exact encoding of the same additive game in a quantum-style latent state with observable expectations computed from the encoded state.

ResidualMLPSurrogate

Learned classical surrogate mapping joint actions to terminal influence distributions in the same state family.

Additive ground-truth dynamics

$$p_{t+1} = \mathrm{entmax}_{1.5}(Mp_t + \Sigma_i B_i a_i)$$

Observable-based payoff

$$u_i(a) = E[z \sim p_T(a)][o_i(z)] - \lambda_i \|a_i\|^2 - \text{penalty}$$

entmax-1.5 replaces softmax because it is sharper, can produce sparse attention patterns, and avoids a trivially smooth response surface that would be too easy for a surrogate to learn.

RIGOROUS STATEMENT

What exact quantum encoding does and does not claim.

Exact statement

For every feasible joint action a, the additive QuantumEncodedGame induces the same terminal influence distribution as the ClassicalGroundTruthGame. Therefore, best responses, regret, and epsilon-equilibrium properties coincide, up to amplitude-estimation discretization error when utilities are evaluated via observable expectations.

p_t → $|\psi_t\rangle$ with $|\psi_t|^2 = p_t$ → Exact additive update preserved → Observable payoff from encoded state

This is an exact strategic encoding result. It shows model equivalence at the distribution level. It does not, by itself, establish end-to-end wall-clock quantum speedup.

EXPERIMENTAL SETUP

All headline metrics are judged against the additive classical ground truth.

Scenarios

Three additive scenarios: shallow_low_conflict, medium_conflict, and deep_high_conflict.

Headline metrics

Accuracy, Regret, and Epsilon are all rechecked against the additive classical ground truth.

Estimation benchmark

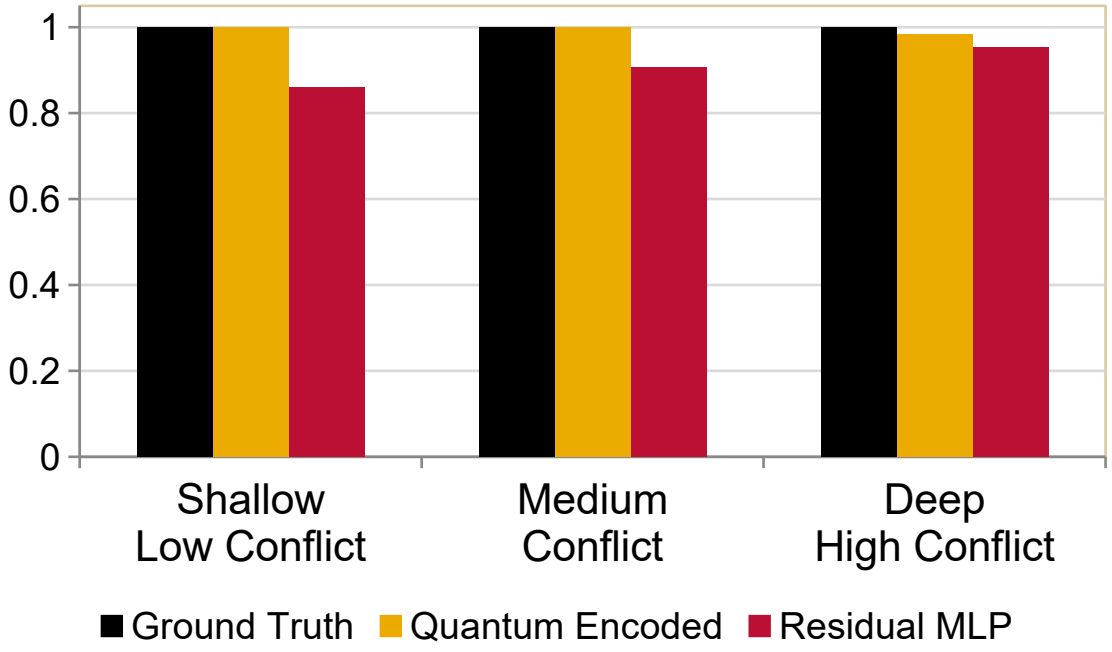
Amplitude Estimation is compared to Monte Carlo on observable-payoff estimation under equal query budgets.

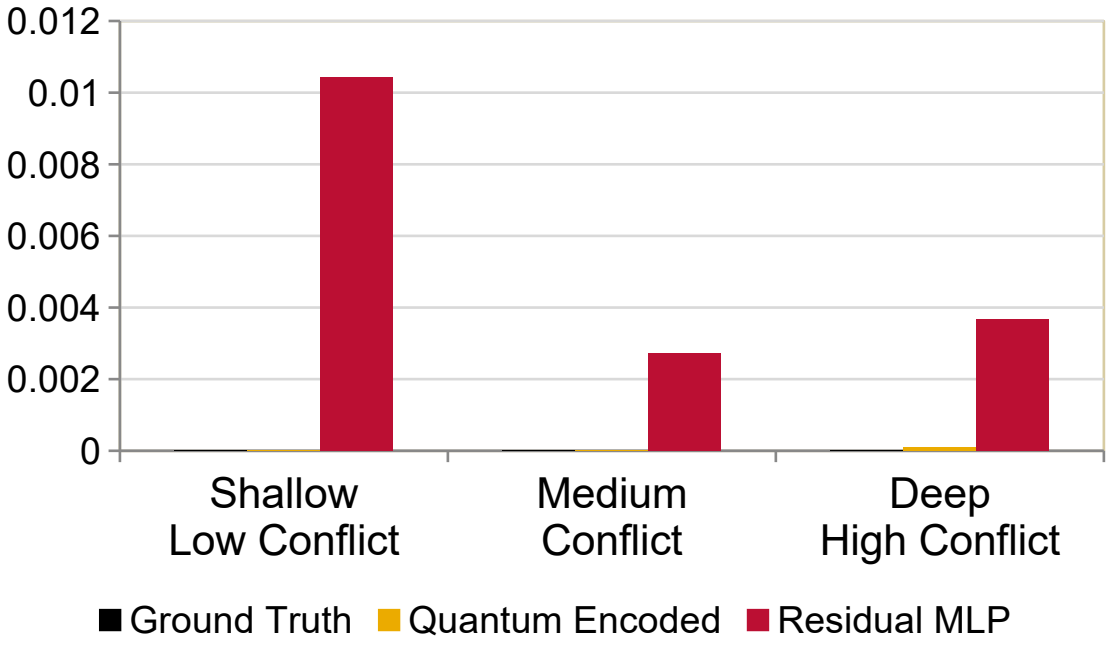
Ground Truth → **Quantum Encoded** → **Residual MLP** → **Evaluate under GT reference**

- Accuracy: whether the approximate model selects the same best-response candidate as the ground truth.
- Regret: exact improvement still available in the ground-truth game after following the candidate recommendation.
- Epsilon: epsilon-Nash slack of the candidate equilibrium when it is checked in the ground-truth game.

STRATEGY AND REGRET RESULTS

Native PowerPoint charts rebuilt from additive report summaries.





Accuracy numbers

Quantum: 1.000 / 1.000 / 0.984. MLP: 0.859 / 0.906 / 0.953.

Regret numbers

Quantum: 8.9e-06 / 0 / 1.0e-04. MLP: 1.04e-02 / 2.72e-03 / 3.66e-03.

Exact quantum encoding preserves strategy structure, while the learned surrogate remains approximate.

EPSILON RESULT AND INTERPRETATION

Epsilon is a structural consistency check, not the only ranking criterion.

Scenario	Ground Truth	Quantum	Residual MLP
Shallow Low Conflict	0.0713	0.0713	0.0000
Medium Conflict	0.0000	0.0000	0.0000
Deep High Conflict	0.0126	0.0126	0.0126

Reading the table

Quantum matches GT because the additive encoding preserves the same terminal distribution.

MLP caveat

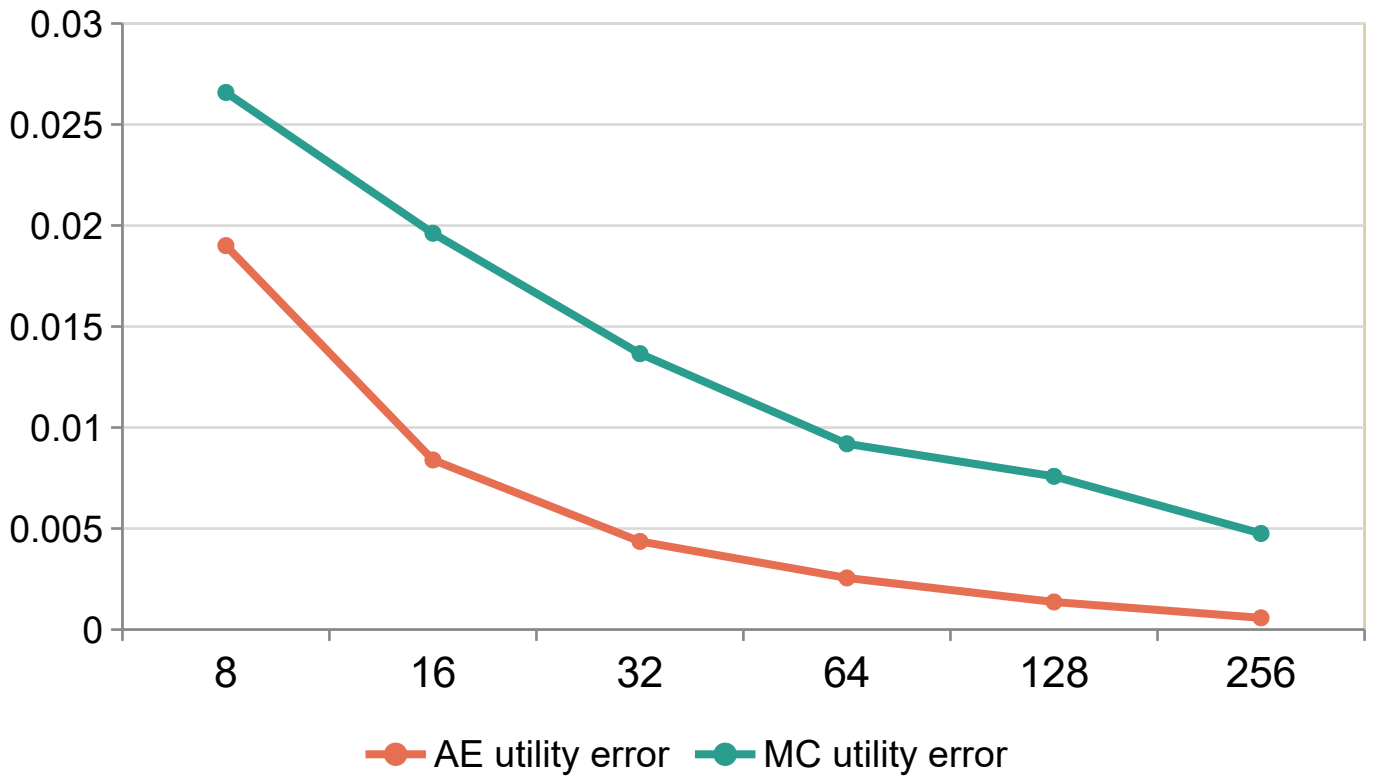
MLP is learned, not exact. A low epsilon in one scenario does not certify structural equivalence.

Epsilon is reported here to check structural consistency. It should be read together with Accuracy and Regret rather than as a standalone ranking metric.

- Quantum and ground truth coincide because they induce the same terminal influence distribution in the additive construction.
- Residual MLP remains useful as a modeling baseline, but it should be interpreted as a learned surrogate rather than as a solver or an exact reformulation.

OBSERVABLE ESTIMATION BENCHMARK

Amplitude Estimation vs Monte Carlo under matched query budgets.



Budget 8

AE 0.0190 vs MC 0.0266

Budget 64

AE 0.00255 vs MC 0.00919

Budget 256

AE 0.000579 vs MC 0.00475

For this observable-estimation task, amplitude estimation achieves lower estimation error than classical sampling at every tested query budget.

WHAT THE RESULTS SUPPORT

A strict claim boundary is part of the contribution.

Supported

1. Exact quantum encoding preserves additive strategic structure.

2. Best responses, regret, and epsilon agree with GT up to AE discretization in observable-based utilities.

3. AE beats Monte Carlo on observable-payoff error under equal query budgets.

Not Supported

1. Universal quantum dominance over classical methods.

2. Hardware-level wall-clock speedup claims.

3. Claims beyond the additive observable-estimation benchmark.

This boundary makes the additive story defensible: exact encoding is a structural statement, while AE provides the measurable performance advantage on a specific observable-estimation task.

CONCLUSION

Final takeaways for the additive-only report.

Takeaway 1

Additive ground truth makes public-opinion shaping explicit and strategically testable.

Takeaway 2

Quantum encoding is exact at the strategic level in this additive construction.

Takeaway 3

Amplitude estimation gives the clean performance advantage in measurable observable-payoff estimation.

Oral closing statement

In the additive setting, the quantum contribution is not a changed game model but an exact strategic encoding plus a cleaner route to observable-payoff estimation.

- This additive deck is intentionally narrower than the full route-2 report and avoids legacy-baseline claims in its main conclusion.
- The rigorous claim is about preserved strategic structure and benchmarked observable estimation, not about universal quantum advantage.